Hypothesis Formulation

10 application-based practice questions



For each scenario: identify the parameter (μ or p), then write H0 and Ha.

Answers are in the speaker notes of each slide.

Q1



Manufacturing

Question 1 of 10

SCENARIO

A dairy plant fills milk pouches labelled 500 ml. After the filling machine was recalibrated last week, the quality inspector suspects it no longer fills 500 ml on average. She picks 40 pouches at random to check.

Write the null and alternative hypotheses

H_0 : _____

H_a : _____

Q2



Food Delivery

SCENARIO

A food delivery app advertises that its average delivery time is at most 30 minutes. A consumer group thinks deliveries actually take longer and tracks 60 orders across the city.

Write the null and alternative hypotheses

H_0 : _____

H_a : _____

Q3



E-commerce

SCENARIO

An e-commerce company claims that no more than 5% of its orders get returned. A warehouse manager feels returns are running higher than that and checks a random sample of 400 orders at $\alpha = 0.05$.

Write the null and alternative hypotheses

H_0 : _____

H_a : _____

Q4



Electronics

SCENARIO

A power bank brand says its product lasts at least 18 hours on a full charge, on average. Reviewers on a tech forum believe the real battery life is shorter. They test 25 units.

Write the null and alternative hypotheses

H_0 : _____

H_a : _____

Q5



Admissions

SCENARIO

Last year, 35% of MBA applicants at a business school picked Marketing as their specialisation. The admissions head believes this share has changed this year and surveys 200 new applicants.

Write the null and alternative hypotheses

H_0 : _____

H_a : _____

Q6

Education

Question 6 of 10

SCENARIO

Students at a coaching centre have historically averaged 62 marks on a mock aptitude test. The centre introduces a new teaching method and claims it will raise the average score. A batch of 36 students is tested at 1% significance.

Write the null and alternative hypotheses

H_0 : _____

H_a : _____

Q7



Healthcare

SCENARIO

A hospital's emergency ward has had an average waiting time of 45 minutes. After rolling out a new triage system, management believes waiting times have gone down. They record the wait for 50 patients.

Write the null and alternative hypotheses

H_0 : _____

H_a : _____

Q8



Banking

Question 8 of 10

SCENARIO

A bank states that 80% of its personal loan applications are processed within 48 hours. Rising customer complaints make the branch manager think the actual proportion is lower. He reviews 150 recent applications.

Write the null and alternative hypotheses

H_0 : _____

H_a : _____

Q9



Smartphones

SCENARIO

A smartphone maker wants to advertise that fewer than 2% of its phones need a repair in the first year. Before the ad goes live, the marketing team wants statistical proof and follows 1,000 buyers for a year.

Write the null and alternative hypotheses

H_0 : _____

H_a : _____

Q10



Digital Marketing

SCENARIO

A brand's Instagram ads have had a click-through rate of 3% for months. The creative team launches a new video ad and believes it will get a higher click-through rate. They track 5,000 impressions.

Write the null and alternative hypotheses

H_0 : _____

H_a : _____